Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/23/2020
PI Name:	Yang, Jiseon Ph.D.		
Project Title:	Developing Predictive Model Systems of Polymicrobial Biofilm Formation and Susceptibility to Chemical Disinfectant: A Longitudinal Study with Implications for Spaceflight Systems Integrity and Health Risks		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 SLOANMOBE2016. Sloan ISS MoBE Postdoctoral Fellowship - Post-Doctoral Fellowships in Space Biology to Study the Microbiome of the ISS as a Built Environment: Using ISS as a Microbiological Observatory
Start Date:	06/01/2017	**End Date:**	06/30/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Wolf, Yolanda	**Contact Phone:**	
Contact Email:	wolf@sloan.org		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2020 per Y. Wolf/Sloan Foundation; original end date was 5/31/2019 (Ed., 4/6/2020) NOTE: End date changed to 12/31/2019 per Y. Wolf/Sloan Foundation; original end date was 5/31/2019 (Ed., 11/18/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nickerson, Cheryl Ph.D. (MENTOR/ The Biodesign Institute, Arizona State University)		
Grant/Contract No.:	Sloan P-2017-9852		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP NOTE: This project is funded by the Alfred P. Sloan Foundation, as a result of a Space Act Agreement between NASA and the Sloan Foundation, for Collaborative Research Proposal for Microbiology of the Built Environment (MoBE). The purpose of this Space Act Agreement is to enhance the ability of NASA and the Sloan Foundation to support scientific research and the dissemination of information about the effect of spaceflight on microbial ecosystems in spacecraft such as the International Space Station (ISS) through sponsorship of postdoctoral fellowships. NASA is providing the isolates for all studies. Understanding the impact of spaceflight on microbial ecosystem dynamics and biofilm formation is critical for spacecraft design, operations, and astronaut health. Biofilms are microbial communities that that adhere to surfaces, causing materials degradation, operational systems failure, and serious infections -- and are resistant to disinfectants/antimicrobials. Biofilms have been reported on spacecraft materials associated with regenerative life support systems, including the International Space Station (ISS) potable water system, which presents a serious threat to onboard systems operation and crew health. The ISS is a built environment that has housed over 200 international crew members and their accompanying microbes since 2000. As such, it is characterized by its own microbiome that has been consistently exposed and adapted to the unique low fluid shear environment of microgravity, which is known to cause unexpected microbial responses, and may present a selective pressure that could shape the adaptation and evolution of these microorganisms. Many ISS microbial isolates are known to be biofilm formers and naturally exist in multispecies ecosystems. While a variety of microbial isolates have been recovered from the ISS potable water over the years, an understanding of how adaptation to spaceflight impacts the formation and function of these organisms as mixed polymicrobial communities in their normal ISS habitat is not known. Likewise, there is extremely limited understanding of how these microbial communities impact spaceflight systems operation and integrity (including regenerative life support systems) and risk to human health. A limited number of spaceflight and RWV (rotating wall vessel) studies have shown that single microbial species form biofilms, with evidence suggesting unique biofilm architecture in microgravity. However, biofilms naturally exist as consortia of multiple microbial species, and there is extremely limited knowledge of how multiple bacterial species form polymicrobial communities in ISS liquid environments, or how these microbial communities present in ISS water systems have changed over time under the selective pressures of the microgravity environment. To begin to address this gap, we propose to develop a series of independent ISS potable water microbial community model systems using predominant bacterial isolates obtained in the same year, and compare how adaptation to the ISS environment in each of these water-microbiome systems has impacted their interactions, ability to form biofilms, and resistance to a commonly used chemical disinfectant aboard the ISS. The archived microbial isolates proposed for use have been collected from the ISS potable water system during three specific years over a six-year period. Microbial isolates from the same year will be simultaneously co-cultured as mixed microbial communities in the NASA Rotating Wall Vessel (RWV) bioreactor as a spaceflight analogue culture system. The proposed methods to accomplish the research objectives include: real-time video (to monitor polymicrobial growth and biofilm formation in RWVs); use of selective media (to monitor viable population dynamics within the polymicrobial communities and susceptibility to chemical disinfectants); and various staining techniques in combination with microscopy (to investigate biofilm formation/architecture). We hypothesize that polymicrobial interactions within the ISS water microbiome cultured in ground-based spaceflight analogue conditions will present distinct differences in biofilm formation, architecture, population dynamics and resistance to disinfectant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of that knowledge to understand microbial risks to systems operation and integrity of potable water systems, as well as to human health. Specifically, this study will utilize the spaceflight analogue platform to broaden our knowledge of 1) how microbial populations cause biofouling of potable water systems, and 2) for the development of new strategies to prevent microbial biofouling, maintain function, and ensure public health.
Task Progress:	[NOTE (Ed., 4/14/2020): PI submitted to Sloan Foundation in March 2019] In order to profile population dynamics of different species of International Space Station (ISS) potable water isolates in the mixed culture over time, we developed selective media for ISS potable water isolates. Intellectual property is in preparation. As proposed, we developed a predictive model system of polymicrobial cultures isolated from ISS potable water using the ground-based spaceflight analogue culture system, the Rotating Wall Vessel (RWV) bioreactor. Using this system, we examined the population dynamics of microbial co-culture of different species isolated from ISS potable water in the same year. Final data from this test will be reported in the final report. We discovered a strong correlation between microbial interaction and date of isolation, thus, providing a possibility to experimentally investigate microbial evolution associated with mixed populational interactions and dynamics influenced by the spaceflight environment. Specifically, our data revealed distinct microbial relationships among the ISS potable water isolates between different species. A spatiotemporal dependence on interactive behaviors was observed (i.e., favorable, antagonistic, neutral), suggesting a possibility for microevolution within the ISS water microbial ecosystem that may be associated with interactive social behavior within the population. In addition, we discovered colony variation in multiple ISS isolates. The distinct patterns observed among the variants for both phenotypic traits and population dynamics suggests possible mechanisms linking social behavior to evolutionary processes. We also tested the impact of silver fluoride (AgF) which is a disinfectant used in the Russian side of the ISS Water Recovery System (and proposed for use in future NASA human spacecraft potable water systems) on mixed-species population and biofilm formation.
Bibliography Type:	Description: (Last Updated: 09/02/2025)

Abstracts for Journals and Proceedings	Yang J, Barrila J, King O, Bruce R, Ott CM, McLean RJC, Nickerson CA. "Multidrug resistant microbial consortia isolated from International Space Station (ISS) potable water: Multi-species interactions, biofilm formation, metabolic characteristics, and hemolytic subpopulations." Poster presented at the 8th ASM Conference on Biofilms, Washington, DC, October 7-11, 2018. Abstracts. 8th ASM Conference on Biofilms, Washington, DC, October 7-11, 2018. , Oct-2018
Books/Book Chapters	Yang J, Thornhill SG, Barrila J, Nickerson CA, Ott CM, McLean RJC. "Microbiology of the built environment in spacecraft used for human flight." in "Microbiology of Atypical Environments. Methods in Microbiology. Vol. 45." Ed. V. Gurtler, J. Trevors. London: Academic Press, 2018. p. 3-26. Available online 31 August 2018. https://doi.org/10.1016/bs.mim.2018.07.002 , Aug-2018